

Filipov's Selection Theorem

1. $F : \Omega \times X \rightarrow \mathcal{P}(Y)$ measurable

2. $F(\omega, \cdot)$ has closed values for all ω

3. For σ -finite measure space $(\Omega, \mathcal{A}, \mu)$



**$\Rightarrow$ There exists a measurable selection $s : \Omega \rightarrow Y$
such that $s(\omega) \in F(\omega)$ for a.e. $\omega \in \Omega$**

*Applications: Existence of solutions to differential inclusions,
integral equations, and optimal control problems*